

SN DOXY 50%
Water Soluble Powder
(FOR VETERINARY USE ONLY)

COMPOSITION

Doxycycline hyclate 500 mg
Exp q.s to 1 g

PRODUCT DESCRIPTION

The finished product is a yellow color powder.

TARGET SPECIES

Pig and Chicken (broiler, pullet, breeder)

PHARMACODYNAMIC

Doxycycline belongs to the group of the tetracycline antibiotics. These antibiotics have a broad spectrum of antimicrobial activity, sharing the same basic structure of polycyclic naphthacenecarboxamide.

Doxycycline is primarily a bacteriostatic drug. It exerts its action by inhibiting the protein synthesis of the bacterial cell. Inhibition of bacterial protein synthesis results in disturbance of all functions necessary for the life of bacteria. Especially cell-division and the formation of the cell wall are impaired.

Doxycycline is a broad-spectrum antibiotic, active against a large number of Gram-positive and Gram-negative, aerobe and anaerobe micro-organisms, Mycoplasmata, Chlamydiae and Rickettsia.

For *Ornithobacterium rhinotracheale* results demonstrate a great variation from high to low susceptibility, depending on the geographical region where isolates came from.

In pig pathogens resistance against doxycycline may also vary; especially susceptibility figures of *A. pleuropneumoniae* may differ from country to country and even farm to farm.

Four resistance mechanisms acquired by microorganisms against tetracyclines in general have been reported: Decreased accumulation of tetracyclines (decreased permeability of the bacterial cell wall and active efflux), protein protection of the bacterial ribosome, enzymatic inactivation of the antibiotic and rRNA mutations (preventing the tetracycline binding to ribosome). Tetracycline resistance

is usually acquired by means of plasmids or other mobile elements (e.g. conjugative transposones). Cross resistance between tetracyclines has also been described. Due to the greater liposolubility and greater facility to pass through cell membranes (in comparison to tetracycline), doxycycline retains a certain degree of efficacy against microorganisms with acquired resistance to tetracyclines.

PHARMACOKINETIC

Doxycycline is absorbed in the stomach and the first part of the duodenum. Compared to the older tetracyclines the absorption of doxycycline is less affected by the presence of bivalent cations in food. Bioavailability in non-fasted pigs is approximately 21%.

Following oral administration at a dose of 12.8 mg/kg body weight, steady state concentrations during medication range between a C_{min} of 0.40 µg/ml in the early morning to a C_{max} of 0.87 µg/ml in the late afternoon in pigs.

Following administration of doxycycline hyclate at an actual dose of 21 mg/kg body weight to chickens mean plasma concentrations above 1 µg/ml were reached within 6 hours and lasted for 6 hours after cessation of medication. From 24 h up to 96 h after start of treatment the doxycycline plasma concentrations exceeded 2 µg/ml. Following administration of doxycycline hyclate at an actual dose of 10 mg/kg body weight steady state plasma concentrations ranged from 0.75 to 0.93 µg/g between 12 and 96 hours after start of medication.

Because doxycycline is highly lipid soluble, it has a good tissue penetration. Respiratory tract tissue: plasma ratios of 1.3 (healthy lungs), 1.9 (pneumonic lungs) and 2.3 (nasal mucosa) have been reported for doxycycline. Plasma protein binding is high (over 90%).

Doxycycline is scarcely metabolised. Doxycycline is primarily excreted with the faeces.

INDICATIONS

Pigs: For the treatment of the clinical signs associated with porcine respiratory disease caused by *Actinobacillus pleuropneumoniae*, *Pasteurella multocida*, and *Mycoplasma hyopneumoniae* susceptible to doxycycline.

Chickens: Where clinical disease is present in the flock, to reduce mortality, morbidity, and clinical signs and to reduce lesions due to Pasteurellosis caused by *Pasteurella multocida* or to reduce morbidity and lesions in respiratory infections caused by *Ornithobacterium rhinotracheale* (ORT).

WARNINGS AND PRECAUTIONS

1. Special warnings for each target species:

None

2. Special precautions for use:

i. Special precautions for use in animals:

Due to likely variability (time, geographical) in susceptibility of bacteria for doxycycline, especially susceptibility of *A. pleuropneumoniae* and *O. rhinotracheale* may differ from country to country and even farm to farm, bacteriological sampling and susceptibility testing are recommended. Use of the product should be based on culture and sensitivity of micro-organisms from diseased cases on farm. If this is not possible, therapy should be based on local (regional, farm level) epidemiological information about susceptibility of the target bacteria. As eradication of the target pathogens may not be achieved, medication should therefore be combined with good management practices, e.g. good hygiene, proper ventilation, no overstocking.

ii. Special precautions to be taken by the person administering the veterinary medicinal product to animals:

If you know you are allergic to the tetracycline class of antibiotics, special care should be taken when handling this product or the medicated solution. During preparation and administration of the medicated drinking water, skin contact with the product and inhalation of dust particles should be avoided. Wear impermeable gloves (e.g. rubber or latex) and an appropriate dust mask (e.g. disposable half-mask respirator conforming to European Standard EN149) when applying the product. In the event of eye or skin contact, rinse the affected area with large amounts of clean water and if irritation occurs, seek medical attention. Wash hands and contaminated skin immediately after handling the product. If you develop symptoms following exposure such as skin rash, you should seek medical advice and show this

warning to the physician. Swelling of the face, lips or eyes, or difficulty with breathing are more serious symptoms and require urgent medical attention. Do not smoke, eat or drink while handling the product. Take measures to avoid producing dust when incorporating the product into water. Avoid direct contact with skin and eyes when handling the product to prevent sensitisation and contact dermatitis.

- Keep medicine out of reach of children.
- Jauhkan ubat dari kanak-kanak

CONTRA-INDICATIONS

Do not use in case of hypersensitivity to the active substance or to any of the excipients.

Do not use in animals with an impaired liver function.

ADMINISTRATION & DOSAGE

Administration orally with the drinking water

The recommended dose in pigs is: 12.5 mg doxycycline hyclate (25 mg product) per kg body weight per day for 4 consecutive days. If no improvement in clinical signs is seen within this time, the diagnosis should be reviewed and treatment changed. In case of severe infections the medication period may be prolonged for a maximum of 8 consecutive days as determined by the attending veterinary surgeon.

The recommended dose in chickens is: 10 mg doxycycline hyclate (20 mg product) per kg body weight per day for 3-4 consecutive days in case of infections caused by *P. multocida* and 20 mg doxycycline hyclate (40 mg product) per kg body weight per day for 3-4 consecutive days in case of infections caused by *O. rhinotracheale*. Based on the dose to be used, and the number and weight of the animals to be treated, the exact daily amount of product can be calculated. The following formula can be used to calculate the concentration of the product in drinking water:

$$\frac{\text{... mg product/ kg b.w/day} \times \text{Mean body weight (kg) of animals to be treated}}{\text{Mean daily water consumption (l) per animal}} = \frac{\text{... mg product}}{\text{per l drinking water}}$$

To ensure a correct dosage body weight should be determined as accurately as possible. The uptake of medicated drinking water depends on the clinical condition of the animals. In order to obtain the correct dosage the concentration of doxycycline has to be adjusted accordingly. The use of suitably calibrated weighing equipment is recommended if part packs are used. The daily amount is to be added to the drinking water such that all medication will be consumed in 24 hours. Medicated drinking water should be refreshed or replaced every 24 hours.

INTERACTION WITH OTHER MEDICAMENTS

Do not combine with antibiotics that are bacteriocidal e.g. penicillins or cephalosporins. Absorption of doxycycline can be decreased in the presence of high quantities of calcium, iron, magnesium or aluminum in the diet. Do not administer together with antacids, kaolin and iron preparations. It is advised that the interval between the administration of other products containing polyvalent cations should be 1-2 hours because they limit the absorption of tetracyclines. Doxycycline increases the action of anticoagulants. The solubility of the product is pH dependent and will precipitate if mixed in alkaline solution. Do not store the drinking water in metallic containers.

PREGNANCY AND LACTATION

Doxycycline has a low affinity for forming complexes with calcium and studies have demonstrated that doxycycline scarcely affects skeleton formation. No negative effects were observed in poultry after the administration of therapeutic doses of doxycycline. In the absence of specific studies the use of the product is not recommended during pregnancy or lactation.

OVERDOSE AND TREATMENT

Overdoses up to 1.6 times the label recommended dose resulted in no clinical signs that could be attributed to treatment. Poultry tolerate double overdoses of doxycycline (40 mg/kg body weight) without any clinical effect.

SIDE EFFECT

Tetracyclines may - in very rare cases - induce photosensitivity and allergic reactions. If suspected adverse reactions occur, treatment should be discontinued. Inform your veterinary surgeon if adverse reactions occur that are not stated.

WITHDRAWAL TIME

Pigs (Meat and offal): 4 days

Chickens (Meat and offal):

- 3 days, following a dose rate of 10 mg/kg body weight for 4 days.
- 12 days, following a dose rate of 20 mg/kg body weight for 4 days.

Eggs: Not authorised for use in laying birds producing eggs for human consumption.

Do not use within 4 weeks of onset of the laying period

SHELF LIFE: 36 months

SHELF LIFE after first opening & reconstitution: 24 hours

STORAGE

Store in a cool dry place ($\leq 25^{\circ}\text{C}$). Strictly avoid light & heat exposure.

PACKAGING

The powder filled 1 kg in aluminum sachets, by automatic filling machine and sealing, and packed in carton box.

DATE OF REVISION: 31 May 2019

Product Registration Holder & Manufactured by:



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